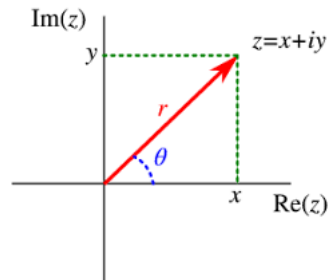


Homework Set 3. First-Order, Second-Order, and Higher-Order Systems

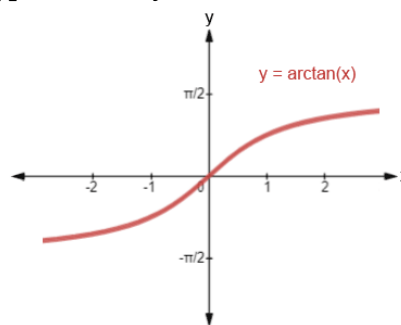
(Due on Friday, February 21, 2025)

1. (15 pt) Calculation with complex numbers.

Recall the Cartesian representation of a complex number $z = x + iy$ and the polar representation $z = re^{i\theta}$. You already know that to convert polar to Cartesian representation, $x = r \cos \theta$ and $y = r \sin \theta$.



- (a) When converting $z = x + yi$ to $z = re^{i\theta}$, obviously we have $r = \sqrt{x^2 + y^2}$. However, the simplistic formula " $\theta = \arctan(y/x)$ " is applicable only if $x > 0$, since " $\arctan$ " is valued in $(-\pi/2, +\pi/2)$.



Therefore, please fill out the correct formula for $-\pi \leq \theta < \pi$:

$$\theta = \begin{cases} \arctan(y/x), & x > 0 \\ ? - \arctan(y/|x|), & x < 0, y > 0 \\ ? - \arctan(y/|x|), & x < 0, y \leq 0 \\ ?, & x = 0, y > 0 \\ ?, & x = 0, y < 0 \end{cases}$$

- (b) Recall from the previous homework: $|z_1 z_2| = |z_1| \cdot |z_2|$, $\angle(z_1 z_2) = \angle z_1 + \angle z_2$, and whenever z_2 is nonzero, $|z_1/z_2| = |z_1|/|z_2|$, $\angle(z_1/z_2) = \angle z_1 - \angle z_2$. In particular, $|z^2| = |z|^2$, $\angle(z^2) = 2\angle z$, $|1/z| = 1/|z|$, $\angle(1/z) = -\angle z$. Please find the **polar representations** of the following numbers (where a, b, c are **positive real** constants):

$$z_1 = \frac{1}{1+ai}, \quad z_2 = \frac{1-ai}{(1+bi)(1+ci)^2}, \quad z_3 = \frac{(1-bi)^3}{1-a^2+2aci} \text{ (assuming } |a| < 1), \quad z_4 = \frac{(1-ai)(1+bi)}{(1+ai)^2} e^{b-ci}.$$

For example, $z_1 = r_1 \exp(i\theta_1)$ where $r_1 = 1/\sqrt{1+a^2}$ and $\theta_1 = -\arctan a$.

2. (15 pt) First-order system. [Seborg 5.2]

A heater for a semiconductor wafer has first-order dynamics, that is, the transfer function relating changes in temperature T to changes in the heater input power level P is

$$\frac{T'(s)}{P'(s)} = \frac{K}{\tau s + 1}$$

where K has units [$^{\circ}\text{C}/\text{kW}$] and τ has units [min]. The process is at steady state when an engineer changes the power input stepwise from 1 to 2 kW. She notes the following:

- The process temperature initially is 100°C .
- 4 minutes after changing the power input, the temperature is 400°C .
- 30 minutes later the temperature is 500°C .

(a) What are K and τ in the process transfer function?

(b) If at another time the engineer changes the power input linearly at a rate of 0.5 kW/min , what can you say about the maximum rate of change of process temperature: When will it occur? How large will it be?

3. (15 pt) **Overdamped second-order system, which can be considered as the parallel connection (sum) of two first-order system.** [Seborg 5.6, part of]:

Consider transfer function

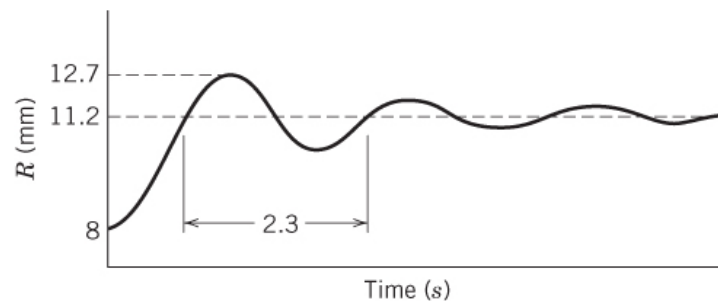
$$G(s) = \frac{Y(s)}{U(s)} = \frac{10}{(5s + 1)(3s + 1)}.$$

What is $y(t \rightarrow \infty)$ for the following inputs:

- (a) Step input of height M ?
- (b) Unit impulse input $\delta(t)$?

4. (15 pt) **Underdamped second-order system.** [Seborg 5.14]

A step change from 15 to 31 psi in actual pressure results in the measured response from a pressure-indicating element shown in the figure below.



(a) Assuming second-order dynamics, calculate all important parameters and write an approximate transfer function in the form

$$\frac{R'(s)}{P'(s)} = \frac{K}{\tau^2 s^2 + 2\zeta\tau s + 1}$$

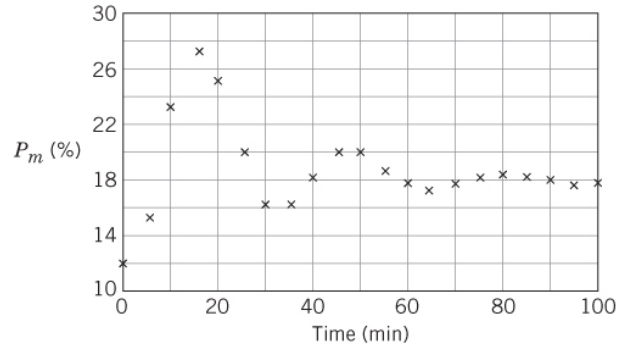
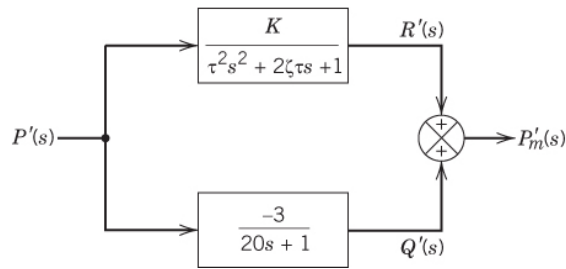
where R' is the instrument output deviation (mm) and P' is the actual pressure deviation (psi).

(b) Write an equivalent differential equation model in terms of actual (not deviation) variables.

5. (20 pt) **High-order dynamics due to connection of multiple simpler low-order models.** [Seborg 6.7]

A pressure-measuring device has been analyzed and can be described by a model with the structure shown in the left figure below. In other words, the device responds to pressure changes as if it were a first-order process in parallel with a second-order process. Preliminary tests have shown that the gain of the first-

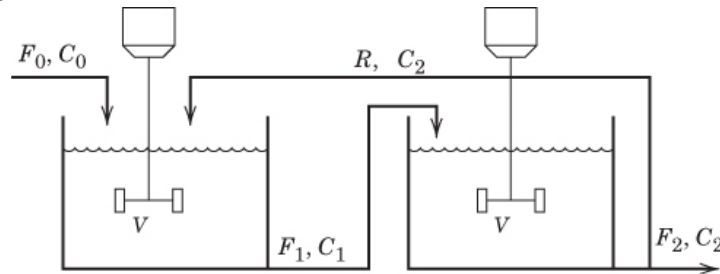
order process is -3 and the time constant equals 20 min. An additional test is made on the overall system. The actual output P_m (not P_m') resulting from a step change in P from 4 to 6 psi is plotted in the right figure below.



- Determine $Q'(s)$.
- Estimate the values of K , τ , and ζ .
- What is the overall transfer function for the measurement device $P_m'(s)/P'(s)$?

6. (20 pt) Second-order dynamics due to recycle connections. [Seborg 6.19]

Consider the following connection of isothermal reactors. A first-order reaction $A \rightarrow B$ occurs in both reactors, with reaction rate constant k . The volumes of liquid in the reactors, V , are constant and equal; the flow rates F_0, F_1, F_2 and R (recycle ratio) are constant. Assume constant physical properties and a negligible time delay for the recycle line.



- Write a mathematical model for this process.
- Derive a transfer function model relating the output concentration of A, C_2 to the inlet concentration of A, C_0 .
- Verify that, in the limit of no recycle ($R \rightarrow 0$), the transfer function derived in (b) is equivalent to the transfer function of the two tanks connected in series.
- Show that when $k = 0$ and a very large recycle flow rate is used (i.e., the limit as $R \rightarrow \infty$), the transfer function derived in (b) becomes the transfer function of a single tank that has the volume equal to $2V$ and a gain of 1.

Hint: Recognize that $F_1 = RF_2 + F_2$ and $F_0 = F_2$.

7. (Bonus: 10 pt) Process control in practice: Identifying a model from step test data. [Seborg 7.3]

In industrial practice, engineers often do not build models from first principles, but instead put a step signal (or other designed signals) on the input and “identify” the model from the resulting output response data. This procedure, called “*system identification*”, can be done either by engineer’s flesh-eye readings, rough calculations, or statistical software. Matlab and Python have well established toolboxes for it. On the other

hand, since system identification is an extremely time-consuming (and hence expensive) task, it remains a focus of today's industrial control technology research and development.

To answer the following question, you may want to read Sections 7.1 and 7.2 of Seborg first.

A process consists of two stirred tanks with input q and outputs T_1 and T_2 (see figure below). To test the hypothesis that the dynamics in each tank are basically first-order, a step change in q is made from 82 to 85 L/min, with output responses given in the table.

```

graph LR
    q --> Tank1[Tank 1]
    Tank1 -- T1 --> Tank2[Tank 2]
    Tank2 -- T2 --> T2_out[T2]

```

$t(\text{min})$	$T_1(^{\circ}\text{C})$	$T_2(^{\circ}\text{C})$	$t(\text{min})$	$T_1(^{\circ}\text{C})$	$T_2(^{\circ}\text{C})$
0	10.00	20.00	11	17.80	25.77
1	12.27	20.65	12	17.85	25.84
2	13.89	21.79	13	17.89	25.88
3	15.06	22.83	14	17.92	25.92
4	15.89	23.68	15	17.95	25.94
5	16.49	24.32	16	17.96	25.96
6	16.91	24.79	17	17.97	25.97
7	17.22	25.13	18	17.98	25.98
8	17.44	25.38	19	17.99	25.98
9	17.60	25.55	20	17.99	25.99
10	17.71	25.68	50	18.00	26.00

- (a) Find the transfer functions $T_1'(s)/Q'(s)$ and $T_2'(s)/T_1'(s)$. Assume that they are of the form $K_i/(\tau_i s + 1)$.
- (b) Calculate the model responses to the same step change in q and plot with the experimental data.

8. (Bonus: 10 pt) Process control in depth: Lead-lag dynamics. [Seborg 6.4]

Consider a process model (for simplicity, you may assume $K = 1$):

$$\frac{Y(s)}{U(s)} = \frac{K(\tau_a s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)} \quad (\tau_1 > \tau_2)$$

For a step input, **prove** that:

- (a) $y(t)$ can exhibit an extremum (maximum or minimum value) in the step response only if

$$\frac{1 - \tau_a/\tau_2}{1 - \tau_a/\tau_1} > 1.$$

- (b) Overshoot occurs only for $\tau_a / \tau_1 > 1$.
- (c) Inverse response occurs (namely an extreme of $y(t)$ has a different sign from K) only for $\tau_a < 0$.
- (d) If an extremum in y exists, the time at which it occurs can be found analytically. What is it?